

Global Seismic Series: Statistical Analysis of Spatiotemporal Clustering in $M \geq 6.5$ Earthquake Catalogs, 1973–2026 CE

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Abstract.

We analyze an analysis catalog of 4,267 unique $M \geq 6.5$ events (modern window 1973–2026: 2,041). A Baiesi-Paczuski η detector with great-circle distance yields 27 algorithmic candidates in the modern window. Catalog-calibrated temporal ETAS (GK mainshocks): mean=27.0, $p_{\text{ETAS}}=1.0$ — N_{obs} consistent with the null. Under the temporal ETAS model used, no anomalous temporal clustering beyond calibrated ETAS; spatial component not modeled. Limitation: spatial component not modeled (Sec. 5.6).

Keywords: global seismicity; seismic series; earthquake clustering; heuristic metric with tectonic hint; Baiesi-Paczuski metric; ETAS validation; Monte Carlo; paleoseismology; remote triggering

1. INTRODUCTION

Large earthquakes do not occur as independent Poisson events. Following the 1992 Landers earthquake (M_w 7.3), Hill et al. [1993] documented remotely triggered seismicity at distances exceeding 1,000 km. Brodsky & Prejean [2006] showed that surface waves can initiate swarms in volcanic systems thousands of kilometres away. The 2004 Sumatra–Andaman earthquake (M_w 9.1) was followed by elevated activity in distant regions [Pollitz et al., 1998; Freed & Lin, 2001].

However, the systematic nature of such correlations remains debated. Michael [2011] found that $M \geq 7$ clustering in 1995–2011 is statistically indistinguishable from random fluctuations. Shearer & Stark [2012] reported no increase in global $M \geq 7$ and $M \geq 8$ rates after the 2004 Sumatra event. Kagan & Jackson [1999] confirmed elevated probability of paired events at short separation without resolving long-range links.

The ETAS model [Ogata, 1988] reproduces regional aftershock clustering but does not encode inter-plate correlations. The Baiesi–Paczuski [2004] metric and Zaliapin–Ben-Zion extensions [2008, 2013] provide objective cluster detection but typically use Euclidean distance, ignoring lithospheric connectivity.

Objective. Test (and if warranted, falsify) the hypothesis that physically meaningful multi-regional global series exist, with primary inference on the modern window (1973–2026), using complementary null tests (permutation vs ETAS) and explicit detector liberalness assessment. Scope. We combine nearest-neighbor clustering with heuristic tectonic hint and ETAS validation; this complements global rate tests (Michael 2011; Shearer & Stark 2012) with a different η -linkage statistic but does not supersede their conclusions.

§1.1 Research question. Do physically meaningful multi-regional global series exist in $M \geq 6.5$ (1973–2026)? (a) Permutation: $p=0.0001$ rejects Poisson times only. (b) ETAS MLE: mean=27.0, $p_{\text{ETAS}}=1.0$. (c) Global-series hypothesis: not tested by temporal-only ETAS; spatial null open. (d) WLS control (App. B): $p=1.0$ — coupling illustration, not primary null.

2. DATA AND METHODS

2.1 Catalog compilation

Magnitude notation. Catalog thresholds and detector gates use $M \geq 6.5$ (USGS ComCat, predominantly M_w); individual events are cited with catalog type where relevant (e.g. M_w 7.3).

Three catalogs were merged: USGS ComCat (1900–2026, primary instrumental); ISC Bulletin (relocated hypocenters); and NOAA NGDC (historical and paleoseismic records pre-1900, 47 events). Duplicates were merged using ± 30 days and ≤ 50 km tolerance; source priority: ISC > USGS > NOAA. After deduplication, the catalog contains 4,267 unique $M \geq 6.5$ events (from 4,418 CSV rows; ~ 151 $M < 6.5$ excluded from clustering). USGS ComCat: 2,088 raw (1973–2026) $\rightarrow$ 2,041 after merge/ISC. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041). `quality_score` is metadata, not an

inclusion filter. Primary analysis set: events 1900–2026 (4,218); 47 pre-1900 records remain in CSV (provenance) but are excluded from the primary detector pipeline and ETAS calibration window (1973–2026 only) — see Appendix A. Final catalog: 4,267 $M \geq 6.5$ events (4,418 CSV rows).

2.1.1 Catalog completeness

Maximum-curvature analysis yields $M_c = 6.55$. Maximum-likelihood b -value from 1,688 events above M_c :

$$b = 0.911 \pm 0.018 \quad (n = 1,688 \text{ events})$$

2.2 Tectonic heuristic (deprecated)

The Bird (2003) tectonic-path heuristic is excluded from primary inference (Appendix); great-circle only. Failure to validate shows metric unsuitability, not evidence against global series.

$$\eta_{ij} = t_{ij} * r_{ij}^{1.6} * 10^{(-b * m_i)}$$

$b=1.0$ — deliberate Baiesi & Paczuski (2004) simplification; catalog $b=0.911 \pm 0.018$ for M_c /completeness and MC null only — not in the η formula. Equal $N=27$ does not prove candidate identity (jaccard=1.0); upstream clusters at $b=0.911$ not re-run (~9.8% label mismatch).

2.3 Analysis pipeline

Raw catalogs → dedup → exclude $M < 6.5$ (~151 NOAA) → GK declustering (primary) → η NN forest → sliding windows → merge overlapping → series criteria → MC/ETAS/FDR. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041) — different algorithms (GK: magnitude-dependent windows; ZBZ: η NN + KDE threshold). At global $M \geq 6.5$ scale ZBZ is permissive (sparse events → high η). GK is sole primary for inference; ZBZ sensitivity only, does not replace GK. If ZBZ were primary, $N=27$ likely unchanged (untested).

Why GK-ZBZ counts differ: not contradictory methods — GK removes short-range fore/aftershocks via fixed windows; ZBZ classifies only 1 event with exceptionally low η at global catalog scale. Under fixed gates GK/ZBZ/none all yield $N=27$ (sensitivity_declustering.json); different mainshock labels (24 vs 1) could matter for cluster analysis.

Algorithm specification (§3.4.1). GK WINDOWS, $T(M)/R(M)$ interpolation, magnitude-descending; aftershocks $[0, T]$, foreshocks $[-T/2, 0]$; haversine. find_nearest_neighbor: causal argmin η , $b=1.0$, GC km. identify_clusters: Union-Find, η_0 KDE. global_series: used[] mask, mean GC > 1500 km; merge 142 → 47 across epochs.

2.4 Clustering and detector criteria

1. GK declustering (primary) → mainshocks for η NN forest.

2. η NN forest: $b=1.0$, $r^{1.6}$; tectonic Bird 2003 ($1.5\times$ GC fallback).
3. Sliding windows 1, 2, 5 yr (1-yr step); merge 142→47.
4. Detector gate: $N\geq 4$; $M\geq 6.5$; mean pairwise GC >1500 km.
5. Flinn-Engdahl zone count — diagnostic only (not admission criterion).

2.5 Primary ETAS null

Primary ETAS null uses catalog-calibrated temporal MLE on GK mainshocks (calibrate_etas_mle.py). WLS calibration scripts are Appendix B reproducibility only.

2.6 Statistical validation

Permutation statement: $p=0.0001$ rejects Poisson event times only (Ogata 1988); does not confirm global series.

Permutation test: $n = 10,000$, $p \leq 0.0001$, $z = -6.17$ (modern). ETAS (primary MLE): mean = 27.0, $p_{\text{ETAS}} = 1.0$, $N = 27$. Interpretation — Sec. 4. GK mainshocks: $N = 27$. BH post-hoc — not discovery.

3. RESULTS

3.1 Identified series

Full historical analysis yields 47 detector candidates (not validated global series): 27 modern ($p \leq 0.0001$), 15 early instrumental ($p = 0.007$; pre-1960 incompleteness caveat), 5 historical candidates (not significant, $p = 0.46$; 47 $M\geq 6.5$ events pre-1900). 142 cluster candidates before filtering. Interpretation — Sec. 4.

Series ID	Events	Regions	M max	Years	Mean lat°	Mean lon°	Notes
1905-1910	193	43	8.8	1905-1910	-60...72	-180...180	Early instrumental; incomplete before ~1960
S047	53	5	8.0	1982-2024	-21.5...18.9	-175.5...155.2	Western Pacific
S170	46	12	9.1	2002-2023	-14.3...33.8	-113.1...167.3	Sumatra 2004 (M 9.1)
S095	25	4	7.9	1989-2017	-8.1...16.5	120.4...146.4	Western Pacific arc
S116	22	5	8.2	1993-2021	-31.7...10.8	-179.5...179.4	South Pacific

Table 1. Top-5 detector candidates (not ETAS-validated physical series).

3.2 Spatial-temporal distribution

Elevated detector candidate frequency (not validated physical series) occurs in 1952–1965 and 2002–2016 (post-Sumatra period). Spatially, clusters concentrate along the circum-Pacific belt (Kamchatka, Kuril Islands, Japan, Tonga, Indonesia). Tectonic diagnostic: median $\Delta\log_{10}\eta = +0.28$ on random pairs (mostly $1.5\times$ GC fallback).

3.3 Consolidated sensitivity table (modern window)

Parameter	Setting	N_series
GC gate	1000 km	27
GC gate	1500 km (baseline)	27
GC gate	2000 km	27
Window	1 yr	53
Window	2 yr (baseline)	27
Window	5 yr	11
Window	10 yr	6
b in η	1.0 (BP 2004)	27
b in η	0.911 (catalog)	27
b overlap (series sets)	Jaccard=1.0; upstream ~9.8%	27
Declustering	GK / ZBZ / none	27 / 27 / 27
min_events (strict)	5 / 6 / 8	27 / 27 / 27
Catalog	GK mainshocks only	27
$\eta_0 \pm 20\%$	not_applied	—

Table 2. *N_series* sensitivity under fixed detector gates (mean GC > 1500 km, $N \geq 4$). Sources: *sensitivity_*.json*.

4. DISCUSSION

WLS negative control (App. B): $p_{\text{ETAS}} = 1.0$, mean = 27 = N_{obs} — detector-calibration coupling; supports negative conclusion; WLS invalid for inference; temporal MLE is primary null.

The detector is liberal: primary MLE ETAS $p_{\text{ETAS}} = 1.0$, mean = 27, $N = 27$. Bird tectonic heuristic excluded; failure shows unsuitability, not evidence against global series. Compatible with Michael (2011) and Shearer & Stark (2012).

5. CONCLUSIONS

Primary ETAS-null (temporal MLE): mean = 27.0, $p_{\text{ETAS}} = 1.0$. No anomalous temporal clustering beyond catalog-calibrated ETAS; spatial linkage among remote events remains open. WLS control: $p = 1.0$ — coupling illustration, not primary null. Permutation rejects Poisson times only (Ogata 1988).

1. Heuristic with tectonic hint: 98% GC fallback — failed hypothesis test.
2. 47 detector candidates indistinguishable from ETAS null; ΔCFS /dynamic stress — future work only.

5.5 ETAS null limitations

Primary null uses temporal MLE on GK mainshocks; WLS — Appendix B (negative control). See Sec. 5.6.

5.6 Limitations

We used temporal ETAS only; the spatial component was not modeled; conclusions are strictly limited to temporal clustering. Rejecting global series as a tested null would require spatial ETAS (Ogata 1998); we do not claim that rejection here.

Limitation	Affected step	Impact on main conclusion
η_0 unverified at global scale	GK/ZBZ identify_clusters	KDE valley $\approx 7.1 \times 10^{-6}$; global_series skips $\eta_0 \rightarrow N=27$; mis-specified η_0 shifts GK/ZBZ labels only
$b = 1.0$ vs 0.911 in η	η upstream	$N_{\text{series}} = 27$ both; cluster labels not re-run
No spatial Ogata MLE	ETAS null	Temporal MLE primary; spatial MLE future work
142 windows + merge	Detector	Main source of liberalness
Calibrated $p_{\text{ETAS}} = 1.0$	ETAS test	Consistent with null; supports negative conclusion

Table 3. Limitation → affected step → impact on main conclusion (Sec. 5.6).

Impact analysis. $N = 27$ is computed by global_series() without η_0 filtering; $b = 1.0$ vs 0.911 leaves $N_{\text{series}} = 27$ under fixed gates, but upstream clusters at $b = 0.911$ were not re-verified. 142 sliding windows dominate detector liberalness.

APPENDIX A. PRE-1900 NOAA RECORDS

Forty-seven fragmentary paleoseismic/historical $M \geq 6.5$ records from NOAA NGDC are retained in data/processed/unified_catalog_full.csv for provenance (not removed). These 47 events are excluded from the primary detector pipeline and ETAS calibration window: pipeline_v2.py and calibrate_etas.py use the modern catalog 1973–2026 only ($N=2,041$). Epoch-stratified counts include pre-1900 descriptively via run_full_historical_analysis.py but do not enter primary significance claims.

APPENDIX B. CATALOG-MATCHED WLS NEGATIVE CONTROL

Reproducibility negative control only — not inference. Catalog-matched WLS (results/etas_calibration.json: $\mu \approx 0.103$, $K \approx 0.495$) yields mean=27.0, $p_{\text{ETAS}}=1.0$ ($n=1000$; multiseed stable). Illustrates detector-calibration coupling; not the primary null.

Component	Method
μ	GK mainshocks / T (closed form)
c, p	Omori MLE, Nelder–Mead on 24 delays
K, α	WLS (numpy.linalg.lstsq) on same 24 GK aftershocks

DATA AND CODE AVAILABILITY

Data and code: github.com/marshalkin-ux/paleoseismic-clustering (docs/data_availability.md). External DOI deposition (Zenodo) deferred — GitHub only.

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